Programming in R

R is a language and environment for statistical computing and graphics. It is a simple and effective programming language, which includes conditionals and loops. Often called GNU S, it is used to statistically explore datasets and to make many graphical displays of data.

R is a highly powerful computer language, an environment and integrated suite of software facilities. The functionality of R can be easily extended via packages. A typical R studio window might have four panes as depicted in Figure 1.

The user can type the commands in Pane 1 and press *Ctrl + Enter* in order to execute the entered command. The output will appear in Pane 2, i.e., the pane with the caption 'Console'. Alternatively, the user might type the command in the Console pane itself and obtain the result on pressing the *Enter* key.

This section dwells on the following: accepting input from the keyboard, generating sequences, and random numbers in R.

scan() can be used for obtaining numeric inputs from the keyboard, as shown below:

```
> x2 <- scan()
```

Now the user can enter the numbers in the Console window and press the *Enter* key instead of entering a number in order to indicate the end of data input. The numbers will be assigned to x2.

Sequence-generating operator

The colon (:) can be used to generate sequences in R, as follows:

```
> x <- 11:20   # the integers in the range 11 to 20 (both
inclusive) will be
         # stored  in x
> x     # print the values stored in x
```

For shuffling the values stored in x, the *sample()* function can be used:

```
> q<-sample(x)
> q
```

The values stored in q might be: 13 14 16 19 17 15 12 11 18 20. The values stored in x remain intact.

seq() can be used in R for generating a sequence, as follows:

```
> z <- seq(from=11,to=30,by=3)
> z
```